A profile-anchored dynamic sensitivity for open-source automated insulin delivery

Initial proposal, with the TDD-based dynamic ISF in AndroidAPS and Trio as the comparator

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Status

This is a first draft for discussion. Nothing described here has run on a device or in a simulator, and no parameter in it was fitted. The candidate is a modelling choice assembled from the findings of documents 5, 6 and 7 in this series, and each choice is marked where it is made.

The June 2026 proposal (document 3) was withdrawn for live dosing on 16 September 2026 and stays withdrawn. This document uses neither its square-root law nor its learned level.

The earlier work did not leave one named model behind, so the reading taken of “the updated model” is set out in the section headed “The candidate”. It can be changed there before anything is built on it.

Glucose is in mg/dL throughout because the code is. 100 mg/dL is 5.6 mmol/L and 200 mg/dL is 11.1 mmol/L.

Summary

The existing equation, 1800 / (TDD × ln(glucose / divisor + 1)), does three jobs at once. Each was examined separately in this series.

Job	How the existing equation does it	What the analysis found	In the candidate
Sets the person’s level	From total daily dose (TDD), a constant and the insulin type	Entered settings already follow TDD between people. The	The entered profile sensitivity

Job	How the existing equation does it	What the analysis found	In the candidate
		level implied changes by 22% with the insulin setting alone	
Moves the level with recent dosing	Blend of 8 hour, 24 hour and 7 day doses	Supported, with an exponent between about -0.65 and -1.0 within a person	Kept, as a bounded ratio to the person's own average
Makes a unit count for more as glucose rises	Logarithm of glucose	No support between 100 and 200 mg/dL away from meals. Open between 200 and 300	Removed above 100 mg/dL, kept below it

For one person with a profile sensitivity of 50 mg/dL per unit, 40 U a day and rapid-acting insulin, at 200 mg/dL, stock AndroidAPS calculates 29.3, Trio's logarithmic mode 41.7 and the candidate 50.0.

The equation is the small part of the change. The larger parts are a check of each person's profile before switching (in AndroidAPS the profile sensitivity has not been used for dosing while dynamic ISF was on) and a measurement of what the removed glucose term was delivering after meals. Both come before a period in which the two values are logged and only the existing one doses.

The comparator

The comparator is the dynamic ISF in stock AndroidAPS (upstream master f77abd9b82, 31 October 2026) and stock Trio (upstream main 0b29cd534, 7 October 2026), read from source on 10 November 2026. oref0 as published has autosens and no TDD-based dynamic ISF, so it has nothing to replace. Loop and iAPS were not read.

One person, worked through

Take the person above with glucose at 200 mg/dL and recent dosing equal to their average.

AndroidAPS takes the 40 U, the glucose and a divisor of 55 for rapid-acting insulin, and calculates $1800 / (40 \times \ln(200 / 55 + 1)) = 29.3$ mg/dL per unit. The profile's 50 does not enter. By that number alone, a correction to 100 mg/dL is 3.41 U.

Trio's logarithmic mode starts from the same equation and expresses it as a ratio of the profile value, with a default adjustment factor of 0.8. The ratio comes out above the upper autosens limit of 1.2 and is held there, which gives 50 / 1.2 = 41.7 and a correction of 2.40 U.

So the same equation returns 29.3 in one system and 41.7 in the other for this person. The difference is Trio's bound around the profile and its default factor.

Across glucose

Values in mg/dL per unit for the same person, from `inv013/candidate.py` (`results/inv013_comparator.md`). AndroidAPS is at an adjustment factor of 100%, Trio at its defaults.

Glucose, mg/dL	AndroidAPS, divisor 55	AndroidAPS, divisor 65	AndroidAPS, divisor 75	Trio logarithmic	Trio sigmoid	Candidate
70	54.8	61.6	68.3	68.5	55.7	63.1
80	50.1	56.1	62.0	62.6	53.7	57.7
100	43.4	48.3	53.1	54.3	50.0	50.0
120	38.9	43.0	47.1	48.6	47.1	50.0
150	34.2	37.6	41.0	42.8	44.3	50.0
200	29.3	32.0	34.6	41.7	42.4	50.0
250	26.3	28.5	30.7	41.7	41.8	50.0
300	24.1	26.1	28.0	41.7	41.7	50.0

Units to bring this person to 100 mg/dL, by the sensitivity alone (insulin on board, predictions and delivery limits are left out):

Starting glucose, mg/dL	AndroidAPS, divisor 55	Trio logarithmic	Candidate
150	1.46	1.17	1.00
200	3.41	2.40	2.00
250	5.71	3.60	3.00
300	8.29	4.80	4.00

This person's profile multiplied by their TDD is 2000. The AndroidAPS equation at 100 mg/dL implies 1737 with a divisor of 55, 1932 with 65 and 2124 with 75, so the candidate's distance from the existing value at 100 mg/dL depends on the person and on their insulin setting. In INV-010 the profile value ran from 0.56 to 1.80 of the dynamic value across six people using Boost.

What the code does

AndroidAPS (`OpenAPSSMBPlugin.calculateRawDynIsf`, `DetermineBasalSMB.kt`):

- The dose term is 0.33 of a weighted eight hour total scaled to a day, 0.34 of the seven day average and 0.33 of the most recent day, multiplied by the adjustment factor (default 100%, range 1 to 300%) and the profile percentage.
- The divisor is 55, 65 or 75 according to the insulin peak time.
- In dynamic mode the dose calculation uses this value in place of the profile sensitivity, so the entered value and its schedule through the day are unused.
- Each predicted glucose in the three prediction arrays gets its own sensitivity from the same equation. The dose is sized on a further value calculated at a glucose chosen by one of three branches.
- With any of the five dose totals missing, the plugin falls back to the profile and raises a notification.

Trio (`DynamicISF.swift`, a Swift port of the `trio-oref` JavaScript):

- Logarithmic mode multiplies the profile sensitivity out of the equation and puts it back, so the profile matters only through the autosens limits (defaults 0.7 and 1.2), which bound the result between profile / 1.2 and profile / 0.7.
- Sigmoid mode is 1 at the profile target, so its level is the profile value. TDD enters only as the ratio of a weighted recent dose to the ten day average.
- The dose term is 0.65 of the last two hours' average and 0.35 of the ten day average by default. The insulin factor is 120 minus the peak time.

Trio's sigmoid mode therefore already has two of the candidate's properties, a profile level and a bounded ratio. It differs from the candidate in what happens above target.

What the evidence supports

Part	Finding	Source	Limit
Level between people	Entered sensitivity against TDD has slope -0.979 [-1.030, -0.934] in 581 people, median product 2067.	Document 6, constants update	The pooled record is 51% children. Entered settings record a decision and most are

Part	Finding	Source	Limit
	Restricted to adults the product is 1915		reached through the 1800 rule
Level for one person	Observational estimates run from 0.30 to 1.12 of the entered value according to construction	Document 7	No construction was validated as a level
Level, by prediction	Overnight, a static 1800 rule missed by 34.1 mg/dL, v1 by 36.7 and the entered setting by 37.3 (1,626 people, trial cohorts). On 89 people using oref, the static profile missed by 20.3 and v1 by 24.6	Document 6, INV-008	The two cohorts disagree on whether the profile or a TDD rule is the better level. INV-008 windows were selected on the later outcome
TDD within a person	Exponent -0.645 (standard error 0.033, 1,313 people). The same estimator on simulated people returns -0.67 for a true -0.50 and -1.02 for a true -1.00	Document 6	Between about -0.65 and -1.0 is supported; the squared term of v2 was best for none of 1,626 people
Glucose, 100 to 200 mg/dL	Six shapes predicted within 0.05 mg/dL of each other against a bar of 0.5 set in advance, the flat one first. Apparent glucose dependence was +0.610 within four hours of a meal and +0.207 at six to nine hours	Document 6	Observational, closed-loop records
Glucose, bias of prediction	On common windows the static profile’s bias stayed between 5.0 and 7.8 mg/dL from 100 to 230 mg/dL. That of v1 ran from +3.7 to -33.2	INV-008	Rescaled loop predictions, outcome-selected windows
Glucose near target	Relative sensitivity measured 0.64 at 90 to 120 mg/dL against 1.00 at 120 to 150.	Document 6	A unit appears to do less near target, which is where the body’s own defence against a fall acts

Part	Finding	Source	Limit
	The existing equation assumes 1.18		
Glucose, 200 to 300 mg/dL	Fall per unit at 300 against 200 mg/dL was 0.66 [0.49, 0.73] in REPLACE-BG (24 people) and 0.73 [0.64, 0.93] under Loop (49 people). The existing equation's own ratio is 0.81 to 0.82	Document 5 section 7, INV-012	No simulated arm reached 300 mg/dL, so this estimator is unchecked there

Two things follow for the design. The level evidence is mixed, and the profile is chosen as the anchor on other grounds, given below. Above 200 mg/dL a decline of about the size the existing equation assumes is not excluded.

The candidate

sensitivity = profile sensitivity now ÷ clamp((recent dose ÷ own average dose)^b) × easing(glucose)

easing(glucose) = ln(P / D + 1) ÷ ln(min(glucose, P) / D + 1)

Worked for the same person at 80 mg/dL, with recent dosing 10% above their average: the profile gives 50, the dose ratio of 1.10 is inside the limits and gives 45.5, and the easing at 80 mg/dL is 1.15, so the value is 52.4. At any glucose from 100 mg/dL upward the easing is 1 and the value is 45.5.

Each part, and why it is this way:

- Where the level comes from: the profile sensitivity for the time of day. This is a modelling choice. The reasons are that the person and their clinic can see it and change it, that Trio's sigmoid mode already works this way, that the existing level moves by 22% with the insulin setting for no physiological reason, and that document 7 found no observational estimate fit to replace an entered value. The prediction evidence does not settle it, so a level of 1800 over the person's average TDD is carried as an evaluation arm.
- How much recent dosing moves it: the system's existing blended dose divided by the average it already holds (seven days in AndroidAPS, ten in Trio), raised to
  1. The default b = 1 is what the existing equation does, which leaves this half of the behaviour unchanged. b = 0.65 is an evaluation arm. The

ratio is held inside the existing autosens limits. Across 960 people the blended dose ran from 0.72 to 1.14 of the person's own average (tenth to ninetieth percentile), which moves the value by 1.39 and 0.88.

- What glucose does below the pivot P (default 100 mg/dL, a modelling choice): the existing logarithm, which raises the value by 15% at 80 mg/dL and 26% at 70. No correction is being given there, so this acts through the predictions: insulin on board is predicted to lower glucose further, which brings a reduced or zero basal rate forward. It runs against the measurement, which shows a unit doing less near target. It is kept because the measured effect is read as the body defending against a fall, and a prediction that relies on that defence errs towards more insulin. An arm without easing is carried.
- What glucose does above the pivot: nothing by default. The existing shape from 200 mg/dL upward is an optional arm, since the 200 to 300 result is open.
- The divisor D stays in the easing only, where 55, 65 and 75 differ by under 0.03 at 70 mg/dL. It could be fixed at one value.
- The adjustment factor has no role. Its job in the existing equation is to move the level, and the profile now does that.

With TDD history missing, the ratio is 1 and the value is the profile with easing. The existing fallback and the candidate then agree above the pivot, so the first week on a new pump does not switch mode.

What has to be added

Component	AndroidAPS today	Trio today	Needed for the candidate
Profile sensitivity in the dynamic path	Present in the profile object, unused in dynamic mode	Used	AndroidAPS: pass it into the calculation
Recent and average dose	Five totals, all required	Two hour and ten day averages	Nothing new collected. Expose the ratio
Equation	One function in the plugin	One function, two modes	One new function per system, with shared reference vectors from `inv013/candidate.py`
Sensitivity at predicted glucose	Three prediction lines	Forecast adjustment in logarithmic mode	Call the new function. Above the

Component	AndroidAPS today	Trio today	Needed for the candidate
			pivot the result no longer changes with predicted glucose
Sensitivity for the dose	Three branches	In the oref port	Call the new function at the same glucose each branch already chooses
Stored value	`variableSens` in the loop result	In the determination	Store the candidate beside it during shadow running
Settings	Adjustment factor, option to adjust sensitivity and target	Mode, two adjustment factors, weighting, limits	A mode selector. No new number is shown to the person in the first version
Profile check on switching	None needed today	None needed today	New. Compare the entered profile with what the existing equation has been calculating for that person, and show the ratio before the mode can be turned on
Tests	`ReplayApsResultsTest` with JavaScript parity assets	`DynamicISFTests`, `DynamicISFEnableTests`	Unit tests on the reference vectors, replay assets for the new mode, and a guard test that the existing mode's output is unchanged
After meals	Covered in part by the glucose term	Covered within the limits	See the next section

Autosens needs a decision in AndroidAPS. INV-010 found under Boost that autosens judged deviations against the profile value while the loop dosed on the dynamic one, so part of its ratio reported the gap between the two. With the candidate the two values coincide above the pivot when the dose ratio is 1. The stock autosens path in dynamic mode was not re-read for this draft.

What the replacement removes

By shape alone the existing equation gives 0.68 of its 100 mg/dL value at 200 mg/dL and 0.56 at 300, so it asks for about 1.5 and 1.8 times the insulin for the same distance from target. The candidate removes that.

The series' reading is that much of this extra insulin has been standing in for carbohydrate still arriving after the absorption model stopped counting it. That is the best-supported explanation and it is not proven. For people who announce meals accurately the removal should cost little. For people who do not announce, or whose meals run long, the glucose term may be doing useful work for a reason other than the one it was designed on, and removing it without a replacement would leave them with less insulin after meals.

The systems already have a route for uncounted carbohydrate: oref's deviation-based unannounced-meal prediction and its dynamic carbohydrate absorption, which are credited here as the right place for this work. Whether they are sufficient without the glucose term is the first thing the evaluation has to answer. Two outcomes are possible and both are acceptable: the candidate goes forward as written, or it goes forward only with the above-200 option or a meal-side addition for people who do not announce.

Evaluation path

Arms: the existing equation as shipped; the profile alone with no dynamic term; the candidate; the candidate without easing; the candidate with the option above 200 mg/dL; the candidate at b = 0.65; and the candidate with its level from 1800 over average TDD.

Stage	What is done	What it decides
1. Replay of the equation	Each arm's value calculated on recorded histories (the INV-008 pipeline), split by time since the last meal and by whether meals were announced	How much insulin the removed term was asking for, in whom and when. Descriptive, no pass rule
2. Prediction	Each arm added to the same-window and overnight comparisons of INV-008 and INV-009	The candidate is carried forward when its typical miss is within 0.5 mg/dL of the best arm, the bar document 6 used
3. Simulation	Paired runs by seed in TimSim, with arms for unannounced meals and for a mis-set profile	A harm rule on time below 54 mg/dL, written before the runs, with thresholds taken from a null arm

Stage	What is done	What it decides
4. Shadow running	Both values calculated on the device each cycle, the existing one doses, both are stored	The distribution of the ratio per person, and the profile check above
5. Use	The developer's own device first, then released lines	Stops on the harm rule

Stages 1 and 2 are analysis and carry the analysis checklist. Stages 4 and 5 change code that can change a dose, so they carry unit and guard tests, the repository's review checklist, a replay against real history and a fresh-context review of the diff.

The way back at every stage is the mode selector: the existing equation stays in the code unchanged and is the default until stage 5 has run. A build that includes the new mode names the previous build.

Open questions

- Whether the level should be the profile or a TDD rule. The two cohorts examined disagree, and a profile that has gone unused under dynamic ISF may be stale.
- Whether sensitivity falls between 200 and 300 mg/dL. Two cohorts suggest it does and the estimator has no simulated check in that range.
- Whether the easing below the pivot should exist, given that it runs against the measurement, and whether the pivot should be 100 or 120 mg/dL or follow the target.
- Whether TimSim's oref adapter runs the existing dynamic ISF closely enough to be the comparator in stage 3. This was not checked.
- How the bolus calculator in each system takes its sensitivity in dynamic mode. This was not read.
- Whether the maintainers of AndroidAPS and Trio want one shared definition. The two existing implementations of one equation differ by 42% for the worked person at 200 mg/dL, which argues for shared reference vectors whatever is decided here.

Reproducing the numbers

`python3 -m inv013.candidate` writes `results/inv013_comparator.md`, which holds every number in the comparator tables and the worked examples. `inv013/tests/test_candidate.py` checks the stock equation against the replay code of INV-008 and the candidate's shape. All other figures are quoted from the documents named beside them.